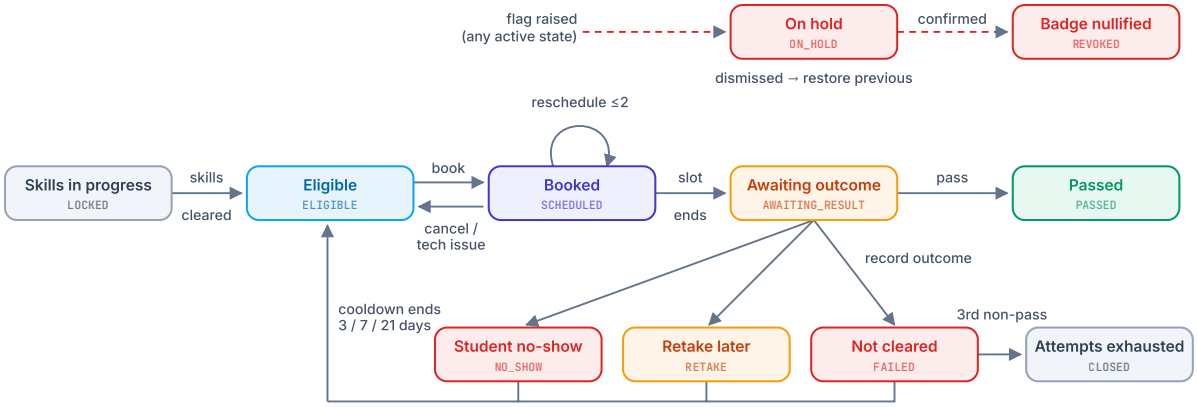
GRIT Level Clearance Interview — Scheduling

AI Product Engineer · Round 1 · Write-up (states, edge cases, data model)

> **Core idea:** every student × badge has exactly one **journey** with one **state**. Every screen — student home, interviewer agenda, ops tracker — is a view of that state, and every change is a validated transition written to an audit log. That's what keeps 15,000 students, a handful of interviewers and a hard Dec 2026 badge deadline consistent.

1 States — what moves a student from one to the next



STATE	STUDENT SEES	ENTERED WHEN	LEAVES WHEN
Skills in progress	Clear your skills first	Journey opens (track's next badge) or a skill is revoked	Last required skill cleared → **Eligible** (unlock message sent)
Eligible · not booked	Ready to book	Skills cleared · cancel · cooldown ended · tech issue · flag dismissed	Student books a slot → **Booked**
Booked	Interview booked	Slot booked (system assigns least-loaded interviewer)	Reschedule (≤2, ≥4h before) · cancel → Eligible · interviewer can't make it → reassigned, else Eligible + priority · slot ends → **Awaiting**
Awaiting outcome	Result on the way	Interview end time passes with no outcome (system job)	Interviewer records: Pass / Retake later / Fail / No-show / Technical issue (24h SLA)
Passed	Badge earned	Outcome = Pass → badge + Miles; becomes Qualiflyer; next badge opens	Only via integrity flag → On hold
Retake later / Not cleared	Not yet / Not cleared	Outcome = Retake (7-day cooldown) or Fail (21-day); feedback mandatory; uses an attempt	Cooldown ends → Eligible · 3rd non-pass → **Attempts exhausted**
Student no-show	Missed interview	Marked ≥10 min after start; 3-day block; 2nd no-show uses an attempt	Block ends → Eligible
On hold	On hold	Ops raises integrity flag from any active state; booking released	Confirmed → **Nullified** · Dismissed → previous state (released booking → Eligible)
Badge nullified	Badge withdrawn	Malpractice confirmed: badge + Miles removed, off the Qualiflyer board	Terminal for the season (appeal is offline)
Attempts exhausted	No attempts left	3 attempts used without a pass	Ops grants an extra attempt (logged) → Eligible

"Live now" (join opens 10 min before) is derived from time, not stored. Pass vs Retake vs Fail differ in consequence, not just label: cooldown length and copy tone.

2 Edge cases & assumptions

Edge cases handled (all demoable in the app)

- **Two students, one slot:** row lock on the slot + partial unique index on `slot_id`; the loser gets "That time was just taken" and a refreshed list.
- **Interviewer over-booked:** per-interviewer daily cap; capacity shown to ops respects it.
- **Interviewer can't make it / takes leave:** booking is first moved to a free colleague at the *same time*; only if none, cancelled with priority rebooking (2h lead instead of 12h), no attempt used.

Assumptions

- 30-min online interview, all times IST; one student per slot.
- Students choose a **time**, not an interviewer — the system assigns the least-loaded qualified interviewer (fairness + load balance).
- Interviewers are qualified per **track**; daily cap per interviewer.
- Eligibility is derived from skill clearances owned by the contest system — this feature never edits them.
- Max 3 attempts per badge per season; cooldowns 7d (retake) / 21d (fail) / 3d (no-show). All in one config file.
- Book ≥12h ahead (interviewer prep), up to 14 days out.

- **Availability edits:** booked slots locked, past read-only; weekly pattern rolls forward, one-off removals stick.
- **Late changes:** reschedule (max 2) / cancel close 4h before.
- **Call drops / no-show:** "Technical issue" uses no attempt; no-show only markable 10 min after start, the 2nd one uses an attempt.
- **Outcome forgotten:** auto "Awaiting outcome" at slot end, 24h SLA, overdue surfaced to ops.
- **Unhelpful feedback:** rubric + ≥40-char student-facing feedback required for non-pass; interviewer sees a live preview of the exact message. Internal notes are separate.
- **Retake bias:** next attempt goes to a different interviewer when one is free; interviewer sees previous attempts' feedback.
- **Malpractice after the badge:** flag → On hold (booking released, badge frozen) → confirm nullifies badge, reverses Miles, re-ranks leaderboard. Dismiss restores.
- **Deadline:** slots never offered past 31 Dec; a cooldown ending after it is flagged "after deadline" for an ops decision.
- **Eligible but silent:** students unbooked 3+ days highlighted; bulk "Nudge".
- **Wrong person acting:** only the assigned interviewer can record; every change is audit-logged with actor and reason.
- Nullification is terminal for the season; appeals happen offline with the campus team.

Questions I'd confirm with the GRIT team

- **Fail vs Retake:** is the only difference cooldown length, or must a failed student re-clear a skill contest?
- **Attempts:** is 3 right? Does a no-show ever count?
- **Deadline:** if a cooldown crosses 31 Dec, do we extend, shorten the cooldown, or let it lapse?
- **Nullification:** can the student re-attempt that badge later in the season? Do downstream badges also fall?
- **Interviewer identity:** show the name before the interview? Avoid same-campus mentors?
- **Interview room:** Meet link vs in-app; record for integrity review?
- **Rubric:** who owns it, and how do we calibrate pass rates across interviewers?

3 Data model

Django models (SQLite locally, Postgres in prod). `badge_journeys.state` is the state machine; `journey_events` is its append-only history.

TABLE	KEY FIELDS	CONSTRAINTS / WHY
`students`	id, roll_no, name, campus_id, batch	Existing GRIT table
`skill_clearances`	student_id, skill_id, level, cleared_at, status (cleared | revoked)	Source of eligibility; owned by contests
`badges`	id, track, level, required_skills jsonb [{skill_id, level}], miles, next_badge_id	Requirements are data, not code
`interviewers`	id, user_id, tracks text[], max_per_day, weekly_pattern jsonb, materialized_until, active	Recurring availability rolled forward by a job
`availability_slots`	id, interviewer_id, start_at timestamptz, end_at, status (open | booked | expired)	UNIQUE(interviewer_id, start_at)
`badge_journeys`	id, student_id, badge_id, **state**, prev_state, eligible_at, attempts_used, extra_attempts, no_shows, cooldown_until, priority, awarded_at, revoked_at, **version**	UNIQUE(student_id, badge_id); CHECK(state in ...); `version` for optimistic locking
`interviews`	id, journey_id, slot_id, interviewer_id, start_at, end_at, attempt_no, status (scheduled | completed | no_show | tech_issue | cancelled), reschedule_count, cancel_reason, meet_link, booked_at	Partial UNIQUE(slot_id) WHERE status='scheduled' → no double booking; partial UNIQUE(journey_id) WHERE status='scheduled' → one live booking per badge
`interview_outcomes`	interview_id PK/FK, result (pass | retake | fail | no_show | tech_issue), scores jsonb, feedback_to_student, internal_notes, recorded_by, recorded_at	CHECK feedback length when result in (retake, fail)
`integrity_flags`	id, journey_id, reason, status (open | confirmed | dismissed), raised_by, raised_at, resolved_by, resolved_at	Reason never shown to student
`notifications`	id, student_id, template_key, payload jsonb, channel (in-app | email | WhatsApp), sent_at, read_at	Copy lives in versioned templates
`journey_events`	id, journey_id, event, from_state, to_state, actor_type, actor_id, detail, created_at	Append-only audit; answers "why is my badge gone?"

- **Booking is one transaction:** lock the slot row (`FOR UPDATE SKIP LOCKED`), insert interview, update journey with a version check, write event + notification. **Jobs** (every minute): slot ended → Awaiting · cooldown ended → Eligible · roll availability forward · SLA reminders.
- **"Who has booked and who hasn't"** is one query over `badge_journeys` grouped by state — no reconciliation sheet.

Built in Django (server-rendered templates, no JavaScript): real logins per role, SQLite/Postgres, DB-level double-booking protection, audit log, 30 automated tests. Exact student messages are in the app and the README.